CIE AS PHYSICS (9702) • QUESTION PAPER

11.1 Atoms, Nuclei & Radiation

Syllabus Section: 11. Particle Physics **Paper Type:** Multiple Choice (Paper 1) **Total:** 35 Questions (35 Marks)

Question 1 Easy [1 mark]

A Geiger-muller tube held at a distance of about 12 cm from a radioactive source. The radiation emitted from the nuclear isotope was entirely stopped by a 2 mm thick sheet of lead.

Which of the statements can be deduced from the above information about the emission from the isotope?

A it could be alpha, beta and gamma radiation

B it could be beta and gamma radiation, but not alpha radiation

C it could be alpha and gamma radiation, but not beta radiation

D it could be alpha and beta radiation, but not gamma radiation

Question 2 Easy [1 mark]

When β^- particles are emitted from identical nuclei in a substance, which of the following remain constant?

A proton number of the nuclei

B nucleon number of the nuclei

C neutron number of the nuclei

D energy of the β-particles

Question 3 Easy [1 mark]

When aluminium $^{27}_{13}Al$ was bombarded with α-particles the first radioactive substance was made. The product of this reaction was an unstable isotope of phosphorus, $^{30}_{15}P$.

Which of the following is a by-product of this reaction?

A a neutron

B a γ-ray

C a β-particle

D an α-particle

Question 4 Easy [1 mark]

The atom $^{133}_{55}Cs$ is a neutral atom

How many protons, neutrons and electrons are in this atom?

	Protons	Neutrons	Electrons
A	133	55	133
B	55	78	55
C	78	55	78
D	55	133	55

A

B

C

D

Question 5 Easy [1 mark]

The following statements are about α-particles.

Which is correct?

A α-particles emitted from a single radioactive isotope have a continuous distribution of energies

B α-particles have less ionising power than β-particles

C the speeds of α-particles can be as high as 1.5×10^{7} m s^{-1}

D the charge of an α-particle is $+1.60 \times 10^{-19}$ C

Question 6 Easy [1 mark]

Which of the following statements about the nucleus of the atom is correct?

A nucleon numbers are unchanged by the emission of β-particles

B in some nuclear processes, mass-energy is not conserved

C for some nuclei, the nucleon number can be less than the proton number

D different isotopic nuclei have different proton numbers

Question 7 Easy [1 mark]

When an α-particle is produced by $^{258}_{92}U$ it produces a new atom X.

What are the values of the proton number and nucleon number for atom X?

	Protons	Nucleons
A	92	256
B	90	254
C	92	250
D	88	254

A

B

C

D

Question 8 Easy [1 mark]

A nucleus Q decays into a nucleus R by emitting an alpha particle followed by two beta particles.

Which statement about this nuclear decay is correct?

A nucleus R is an isotope of nucleus Q

B nucleus R has the same nucleon number as nucleus Q

C beta particle decay occurs when a proton changes into a neutron

D the total mass of the products is equal to the mass of the initial nucleus Q

Question 9 Easy [1 mark]

Four nuclei are represented below

$$^{24}_{13}Q \qquad ^{28}_{14}E \qquad ^{25}_{15}G \qquad ^{25}_{12}M$$

Which of the following statements about these nuclei is correct?

A an uncharged atom of element Q has 24 orbital electrons

B when E absorbs a neutron and then emits an alpha particle, nucleus E transforms into M

C nuclei G and M are isotopes of the same element

D nucleus M could transform into Q by emitting a beta particle

Question 10 Easy [1 mark]

Plutonium-239 $^{239}_{94}Pu$ decays by emitting α-radiation.

Which nuclide is formed from one of these decay reactions? (The product nuclides are represented by X.)

A $^{239}_{95}X$

B $^{237}_{92}X$

C $^{239}_{93}X$

D $^{235}_{92}X$

Question 11 Easy [1 mark]

One of the rows in the table shows a pair of nuclei that are isotopes of one another.

Which row is it?

	Nucleon number	Number of neutrons
A	186 180	112 118
B	184 187	110 110
C	186 182	112 108
D	186 186	110 112

A

B

C

D

Question 12 Easy [1 mark]

Which statement about alpha, beta and gamma radiation is correct?

A gamma radiation has the greatest ionising power

B beta radiation has the greatest ionising power

C alpha radiation has the greatest ionising power

D alpha, beta and gamma radiation have nearly equal ionising powers

Question 13 Easy [1 mark]

Alpha, beta and gamma radiations have various depths of penetration in matter and different charges.

Which row in the table best shows the penetration and charge of each radiation?

	Alpha	Beta	Gamma
A	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium no charge	not fully absorbed by several mm of lead no charge
B	absorbed by several mm of aluminium positive charge	not fully absorbed by several mm of lead negative charge	absorbed by a sheet of card no charge
C	absorbed by a sheet of card negative charge	absorbed by several mm of aluminium positive charge	not fully absorbed by several mm of lead no charge
D	absorbed by a sheet of card positive charge	absorbed by several mm of aluminium negative charge	not fully absorbed by several mm of lead no charge

A

B

C

D

Question 14 Easy [1 mark]

Most α-particles pass straight through when fired at a thin sheet of metal foil, but a few are deflected by a large angle.

Which change would increase the **proportion** of α-particles deflected by a large angle?

A using a double thickness foil

B using a foil made of a metal with fewer protons in its nuclei

C using α-particles with greater kinetic energy

D using an alpha source with a higher activity

Question 15 Easy [1 mark]

New insights were discovered when α-particles were fired at a sheet of gold foil.

Which conclusion can be drawn from the results of the experiment?

A the atomic nucleus contains protons and neutrons

B electrons orbit the atomic nucleus

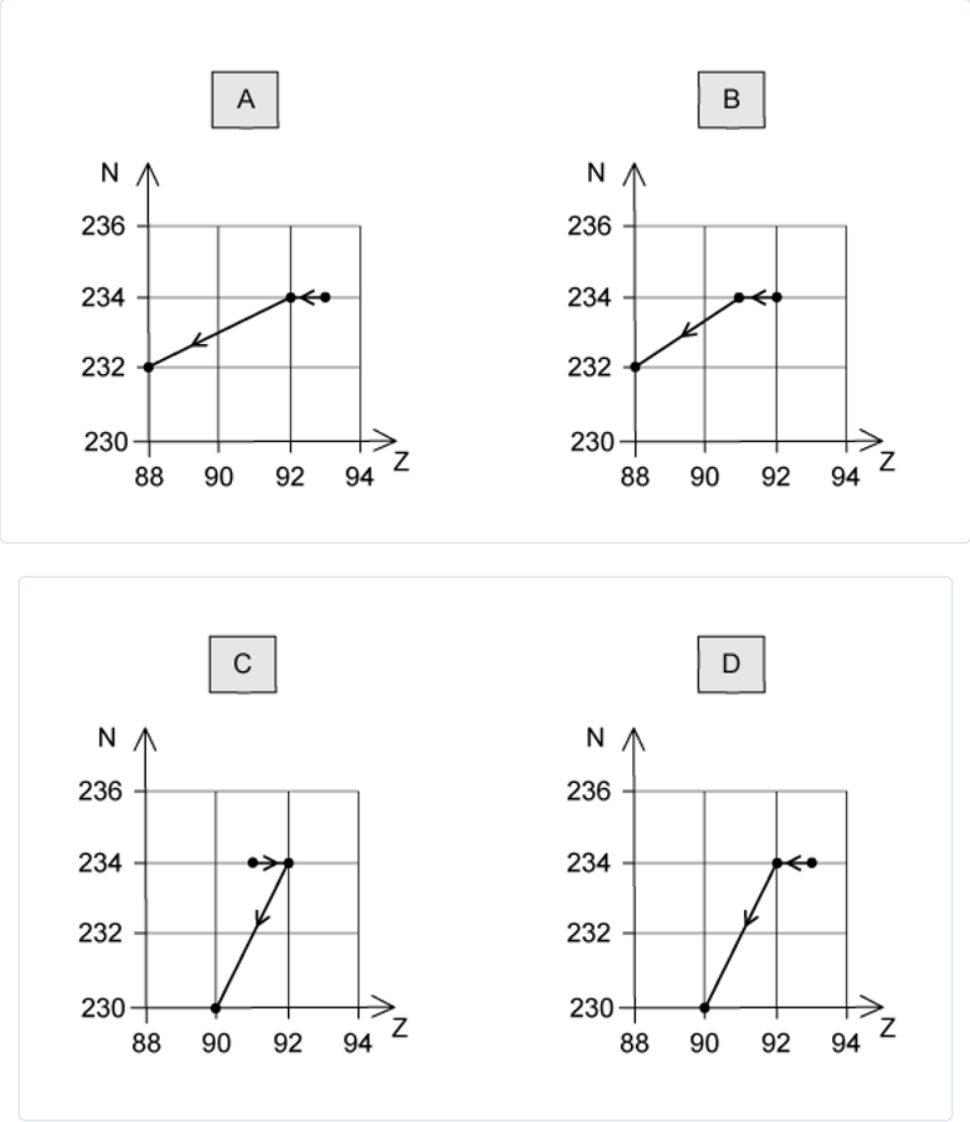
C some atoms of the same element contain different numbers of neutrons

D atomic nuclei occupy a very small fraction of the volume of an atom

Question 16 Medium [1 mark]

A radioactive nucleus is formed by β-decay. This nucleus then decays by α-emission.

The graphs below show the nucleon number N plotted against proton number Z. Which one shows the β-decay followed by the α-emission?



A

B

C

D

Question 17 Medium [1 mark]

Nucleus X decays in two stages to produce nucleus Y.

Which decay sequence will result in the highest number of neutrons in nucleus Y ?

A an α-particle followed by a β-particle

B a β-particle followed by a γ-ray

C a β-particle followed by another β-particle

D an α-particle followed by a γ-ray

Question 18 Medium [1 mark]

The isotope $^{222}_{86}Rn$ decays in a sequence of emissions to form the isotope $^{206}_{82}Pb$.

It will either emit an α-particle or a β-particle at each stage of the decay sequence.

What is the number of stages in the decay sequence?

A 20

B 16

C 8

D 4

Question 19 Medium [1 mark]

An element emits an alpha particle from its radioactive nucleus.

The daughter nucleus then emits a beta particle, and then the daughter nucleus of that reaction emits another beta particle.

Which statement describes the final nuclide that is formed ?

A it is a nuclide of the same element but with different proton number

B it is a nuclide of a different element of higher proton number

C it is a different isotope of the original element

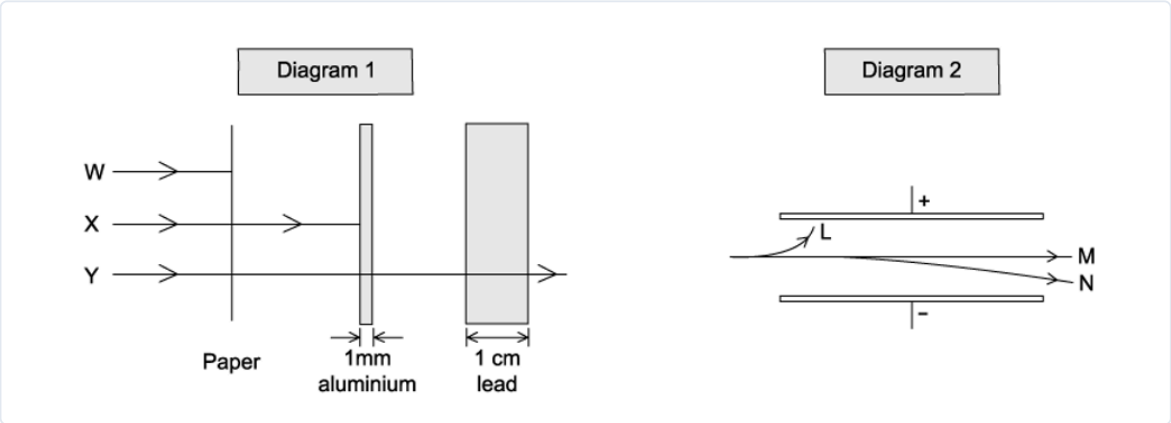
D it is identical to the original nuclide

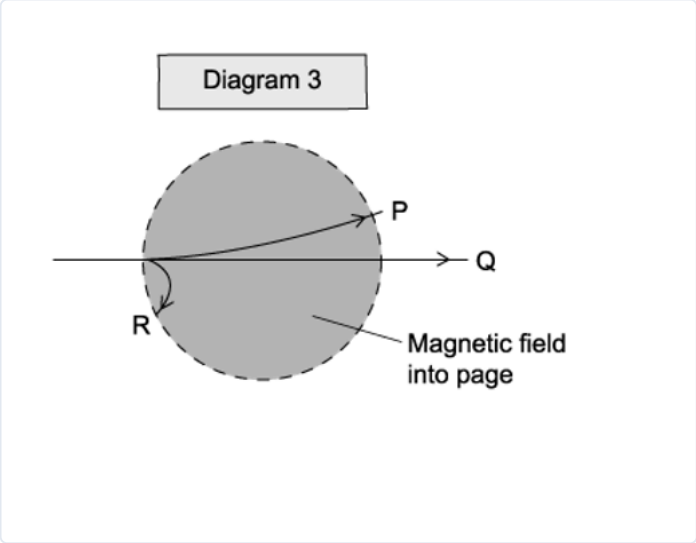
Question 20 Medium [1 mark]

Alpha, beta and gamma radiations:

1	are absorbed to different extents in solids
2	behave differently in an electric field
3	behave differently in a magnetic field

Diagrams 1, 2 and 3 illustrate these behaviours .





Which three labels on these diagrams refer to the **same** kind of radiation?

A X, L, R

B W. L, R

C W, L, P

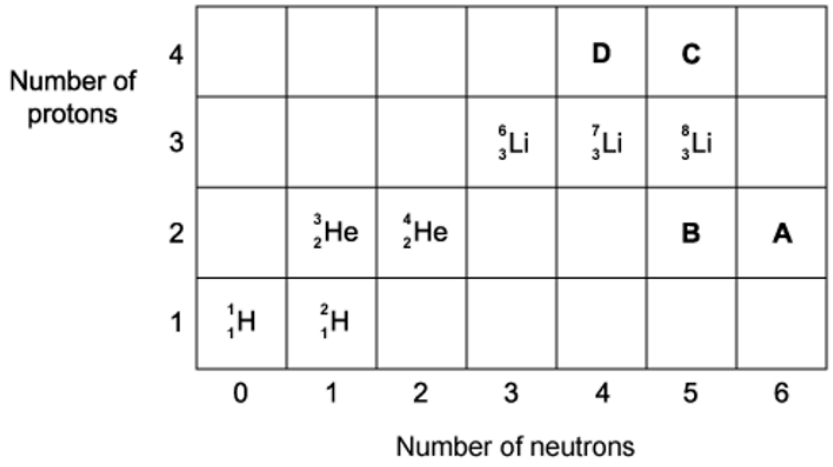
D Y, M, P

Question 21 Medium [1 mark]

The nuclides shown in the grid below are arranged according to the number of protons and neutrons in each.

A nucleus of the nuclide $^{8}_{3}Li$ decays by emitting a β-particle.

What is the resulting nuclide ?



A

B

C

D

Question 22 Medium [1 mark]

A thorium isotope has a nucleon number of 232 and a proton number of 90. It decays to form another isotope of with a nucleon number of 228.

How many alpha particles and beta particles are emitted during this decay?

	Alpha particles	Beta particles
A	0	4
B	1	2
C	1	1
D	2	1

A

B

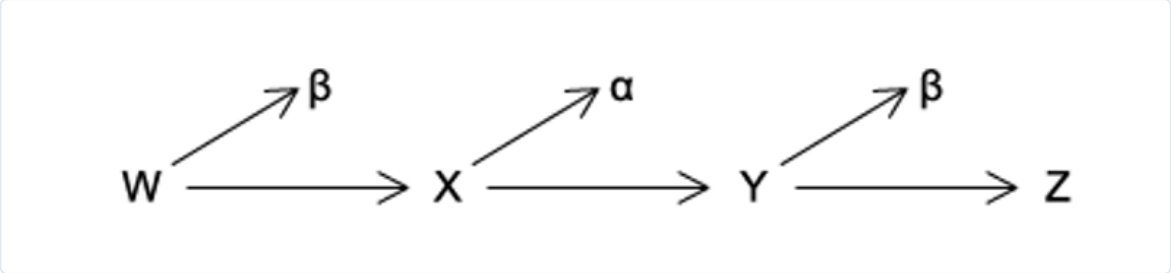
C

D

Question 23 Medium [1 mark]

Three successive radioactive decays are shown in the diagram below; each one results in a particle being emitted.

The first decay results in the emission of a β-particle. The second decay results in the emission of an α-particle. The third decay results in the emission of another β-particle.



Nuclides W and Z are compared. Which statement is correct ?

A W and Z are isotopes of the same element

B Z is a different element of reduced mass

C Z is a different element of lower atomic number

D W and Z are identical in all respects

Question 24 Medium [1 mark]

Antimatter is a particle that is an antiparticle to the corresponding particle. A positron is the antiparticle of an electron.

What is the difference between a positron and an electron ?

A Mass

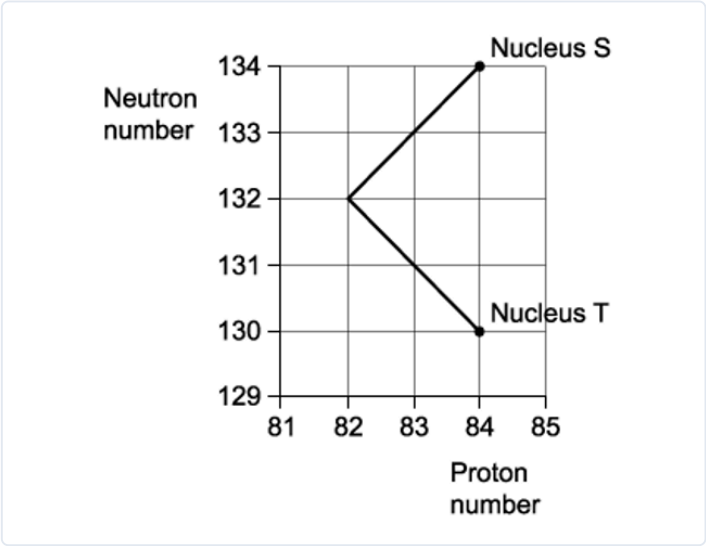
B magnitude of charge

C Charge

D Spin

Question 25 Medium [1 mark]

A sequence of radioactive decays is shown in the graph of neutron number against proton number.



Nucleus S is at the start of the sequence and, after the decays have occurred, nucleus T is formed.

What is emitted during the sequence of decays?

A one α-particle followed by one β-particle

B two β-particles followed by one α-particle

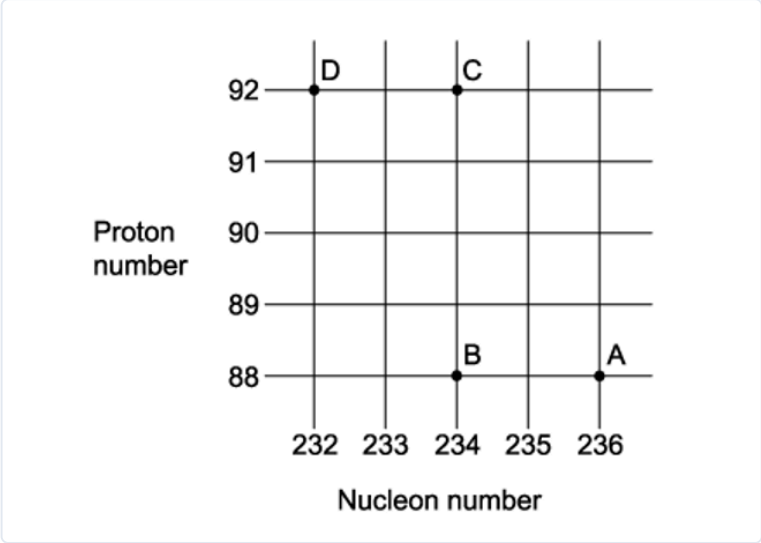
C two α-particles followed by two β-particles

D one α-particle followed by two β-particles

Question 26 Medium [1 mark]

A radioactive substance with a nucleon number of 234 and a proton number of 90, decays by β-emission into a daughter product which in turn decays by further β-emission into a granddaughter product.

Which letter in the diagram represents the granddaughter product ?



A

B

C

D

Question 27 Medium [1 mark]

Astatine is a radioactive substance; it has a nucleon number of 218 and a proton number of 85. When it decays it forms a polonium nucleus, emitting a β^- particle and an α-particle.

What are the nucleon number and the proton number of the polonium nucleus?

	Nucleon number	Proton number
A	214	84
B	214	83
C	216	83
D	215	82

A

B

C

D

Question 28 Medium [1 mark]

An element with an unstable nucleus decays by emitting an alpha particle to become the nucleus of a different element.

The nucleus of the new element is unstable and will emit either an α-particle or a β^- particle. This process continues until an isotope of the original element is formed.

What is the minimum possible number of the particles emitted ?

A 5

B 4

C 3

D 2

Question 29 Medium [1 mark]

When α-particles are directed at gold leaf

1	almost all α-particles pass through without deflection,
2	a few α-particles deviate through large angles.

What are the reasons for these effects?

	1	2
A	the positive charge in an atom is not concentrated enough to deflect an α-particle	occasionally an α-particle experiences many small deflections in the same direction
B	the gold nucleus is very small so most α-particles miss all nuclei	occasionally the path of an α-particle is close to a nucleus
C	most α-particles miss all gold atoms	a few α-particles bounce off gold atoms
D	most α-particles have enough energy to pass right through the gold leaf	gold is very dense so a few low energy α-particles bounce back from the gold surface

A

B

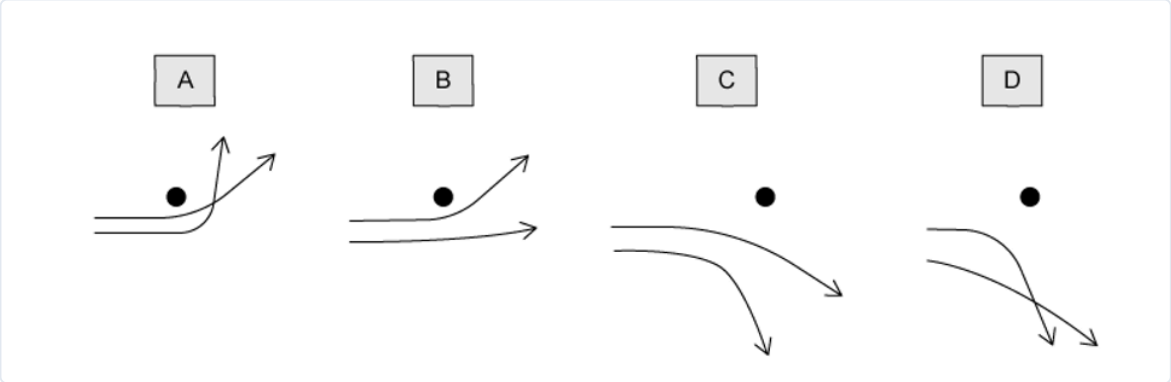
C

D

Question 30 Medium [1 mark]

Two α-particles with equal energies are deflected by a gold nucleus.

Which diagram best represents their paths ?



A

B

C

D

Question 31 Hard [1 mark]

A stationary decay $^{210}_{84}Po$ of by α-emission has a total kinetic energy of E_k

$$^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + \alpha$$

What is the kinetic energy of the daughter nucleus $^{206}_{82}Pb$?

A 0

B between 0 and $\frac{E_k}{2}$

C between $\frac{E_k}{2}$ and E_k

D E_k

Question 32 Hard [1 mark]

Which of the following equations correctly shows an α-particle causing a nuclear reaction ?

A $^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}n$

B $^{17}_{8}O + {}^{4}_{2}He \rightarrow {}^{20}_{9}F + {}^{1}_{1}P$

C $^{17}_{8}O + {}^{0}_{-1}e \rightarrow {}^{13}_{5}B + {}^{4}_{2}He$

D $^{14}_{7}N + {}^{1}_{1}p \rightarrow {}^{11}_{6}C + {}^{4}_{2}He$

Question 33 Hard [1 mark]

Thorium $^{232}_{90}Th$ decays through a series of transformations. The particles emitted in successive transformations are

$$\alpha \quad \beta \quad \beta \quad \gamma \quad \alpha$$

The resulting nuclide may be represented by

A $^{230}_{82}Pb$

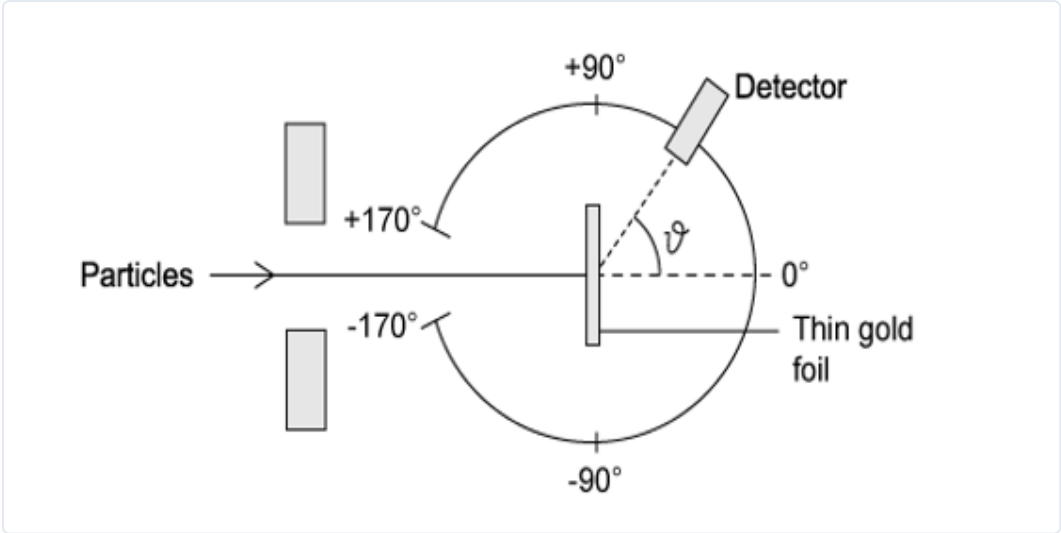
B $^{227}_{85}At$

C $^{225}_{87}Fr$

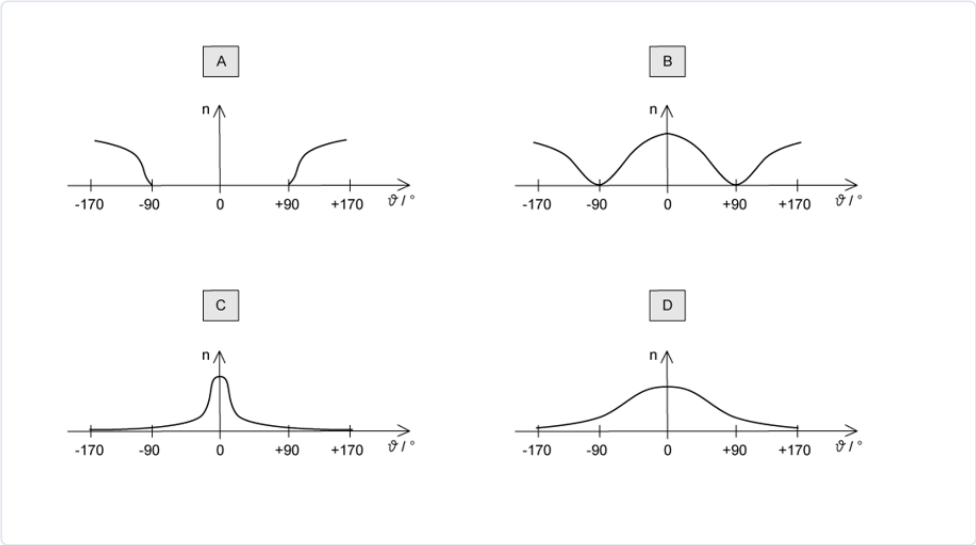
D $^{224}_{88}Ra$

Question 34 Hard [1 mark]

In an α-particle scattering experiment, a student set up the apparatus below to determine the number n of α-particles incident per unit time on a detector held at various angles θ.



Which of the following graph best represents the variation of n with θ ?



A

B

C

D

Question 35 Hard [1 mark]

An unknown isotope ${}^{A}_{Z}X$ decays through a series of transformations into Thorium-228. The atomic number of thorium is 90.

The successive transformations are:

$$\gamma \quad \beta^{-} \quad \alpha \quad \alpha \quad \gamma$$

Which row states the correct isotope name, atomic number and mass number of the unknown isotope ${}^{A}_{Z}X$?

	Isotope name	Atomic number / Z	Mass number / A
A.	Neptunium	93	235
B.	Plutonium	94	237
C.	Neptunium	93	236
D.	Uranium	92	232

A

B

C

D